



Report

Substitution Key

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### key strength of simple substitution random

$$26! = 4,03 \cdot 10^{26}$$

$$1024 \rightarrow 2^{10}$$

$$2^{10} = 10^3$$

$$10^{26} = 10^3 \cdot 10^{23}$$

$$= 100 \cdot (10^3)^8$$

$$= 100 \cdot (2^{10})^8$$

$$= 100 \cdot 2^{80}$$

$$100 = 2^7$$

$$= 2^7 \cdot 2^{80}$$

$$26! = 4.03 \cdot 2^{87}$$

$$= 2^{89}$$